

AN ANALYSIS OF THE HOUSE SALES PRICES IN KING COUNTY

for

MISS AMY WILLIAMS



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KEVIN DIETRICH

2025-03-17

GETTING FAMILIAR WITH THE DATA SET

HYPOTHESIS #1

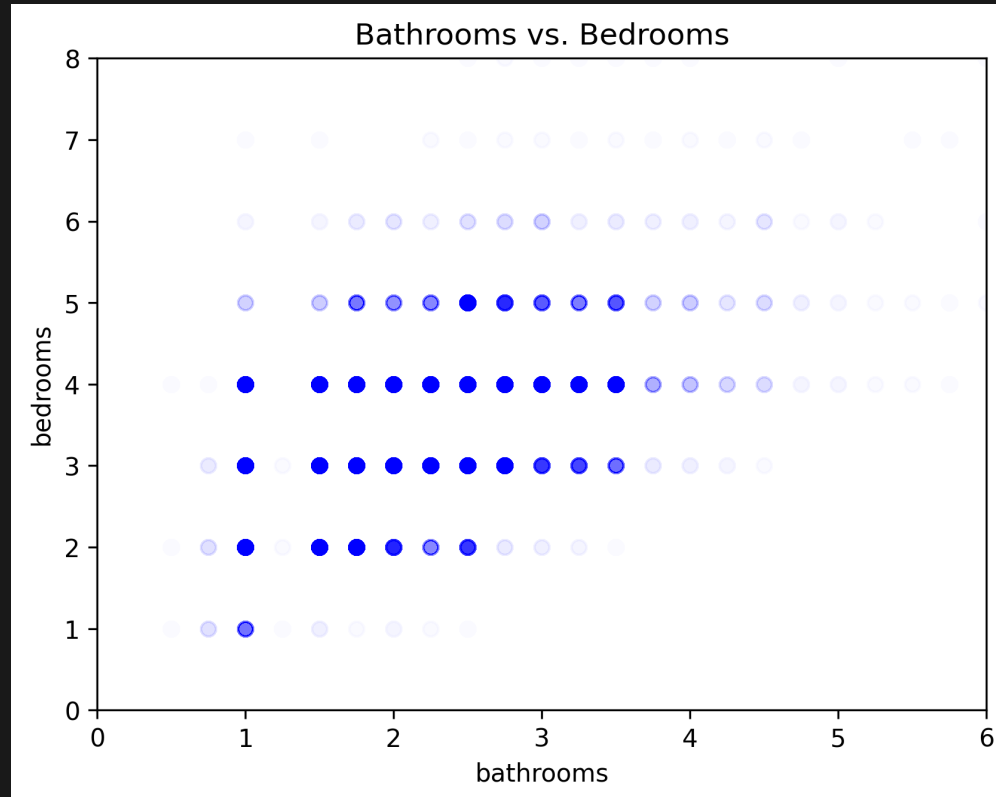
The more bathrooms, the more bedrooms.

Is it true???

The more bathrooms, the more bedrooms?

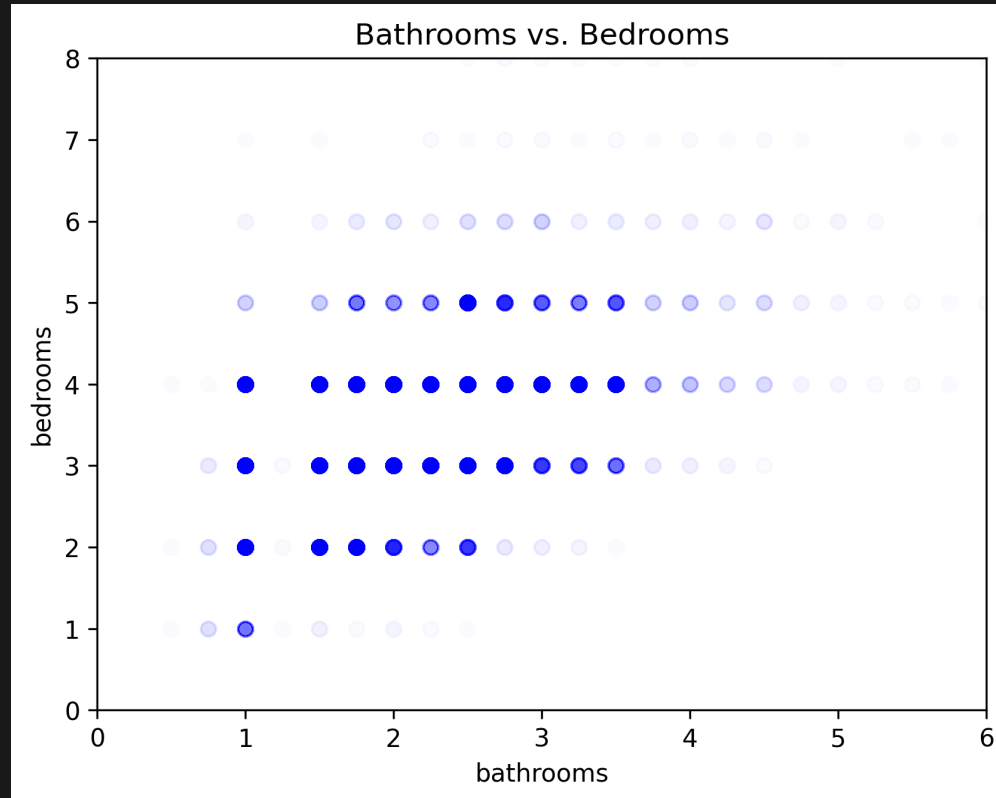
The more bathrooms, the more bedrooms?

Yes!



The more bathrooms, the more bedrooms?

Yes!



Correlation ≈ 0.5

HYPOTHESIS #2

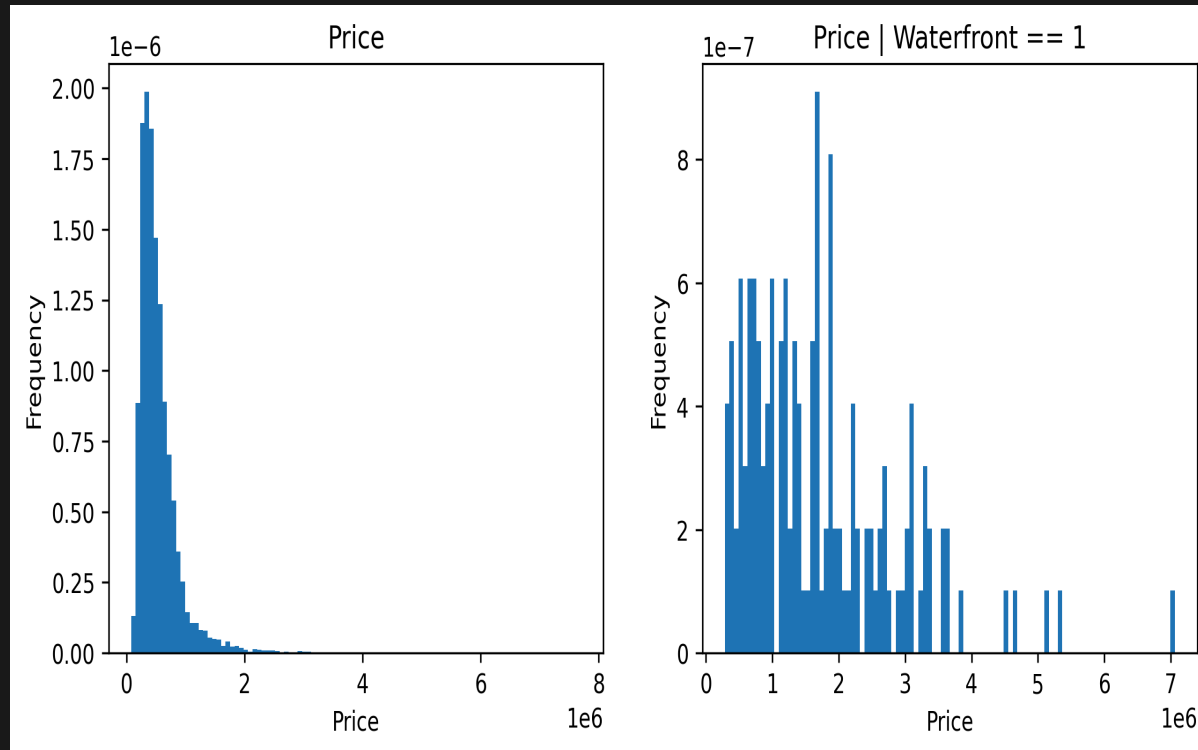
The presence of a waterfront affects the prize.

Is it true???

The presence of a waterfront affects the prize?

The presence of a waterfront affects the prize?

Yes!



HYPOTHESIS #3

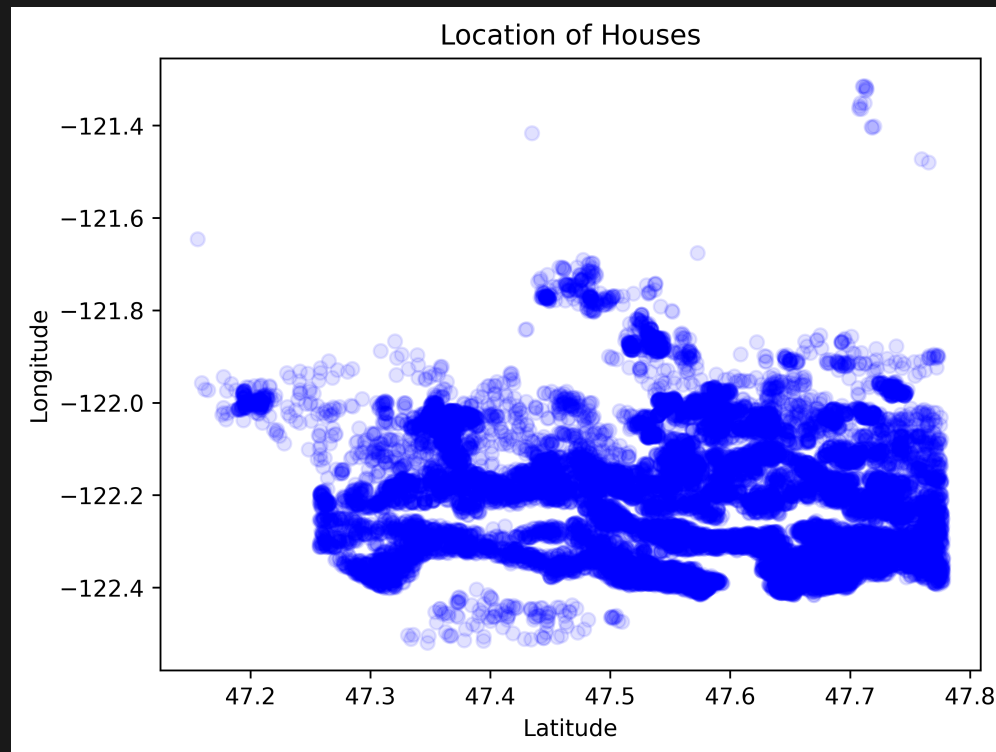
The latitude and the longitude of a house location do not affect each other.

Is it true???

The latitude and the longitude of a house location do not affect each other?

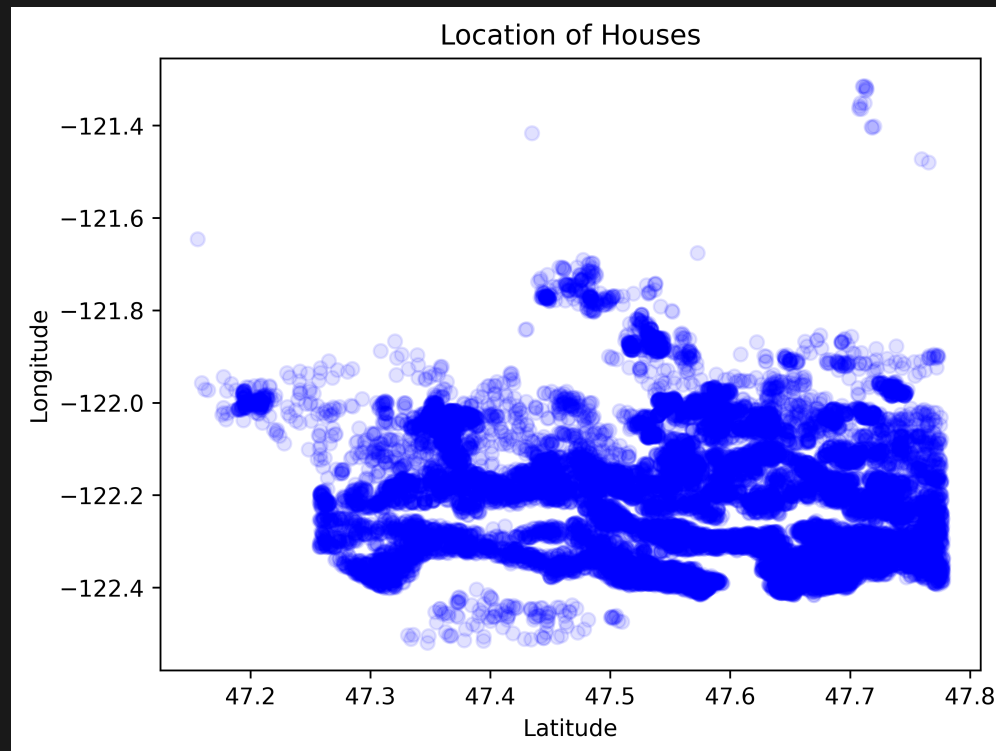
The latitude and the longitude of a house location do not affect each other?

Probably Yes!



The latitude and the longitude of a house location do not affect each other?

Probably Yes!



Correlation ≈ -0.1

OUR CLIENT

Amy Williams, Seller, ~~Mafiosa~~ **businesswoman by trade**, sells several central houses (top 10%) over time, needs average outskirts houses over time to ~~hide from the FBI~~ **avoid entanglements with the FBI.**

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- (top 10%) **centrality**?

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- **exceptionality** (on the outskirts)?

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⇒ **HOW TO DEFINE ...**

- (top 10%) **centrality**?
- **exceptionality** (on the outskirts)?
- **privacy** (on the outskirts)?

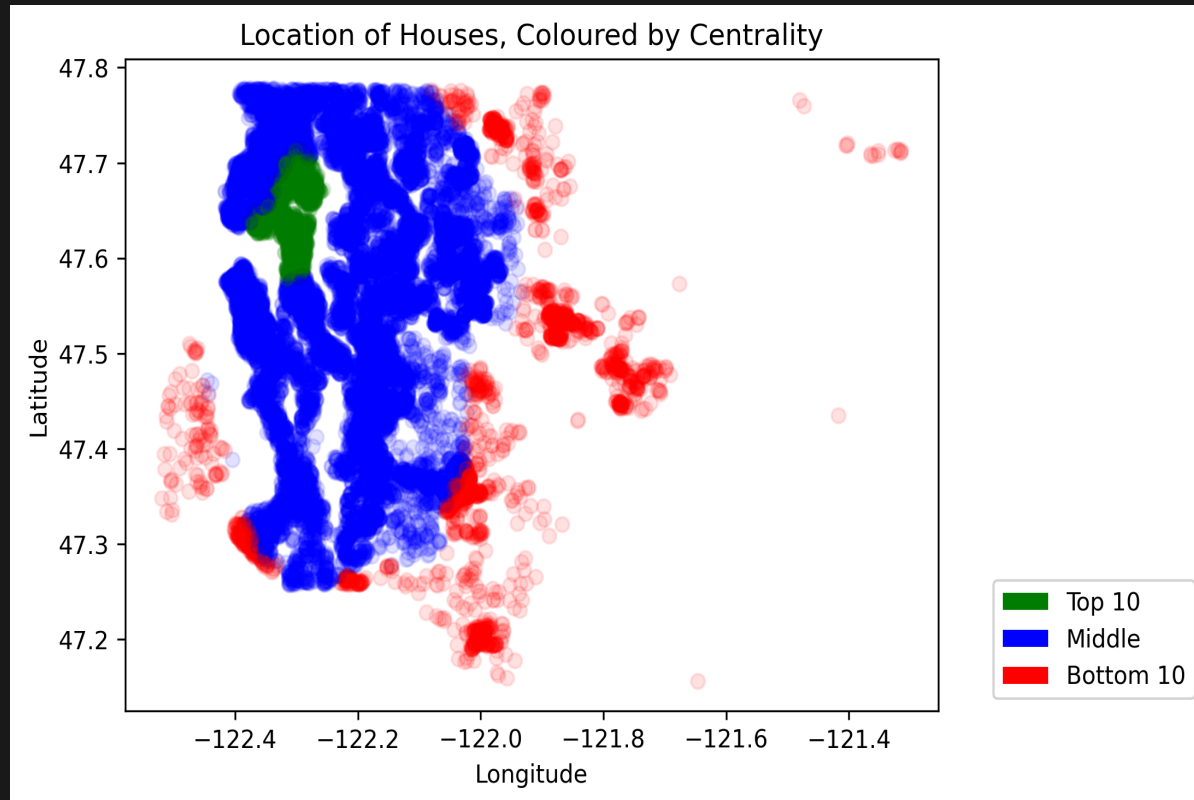
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(TOP 10%) CENTRALITY?

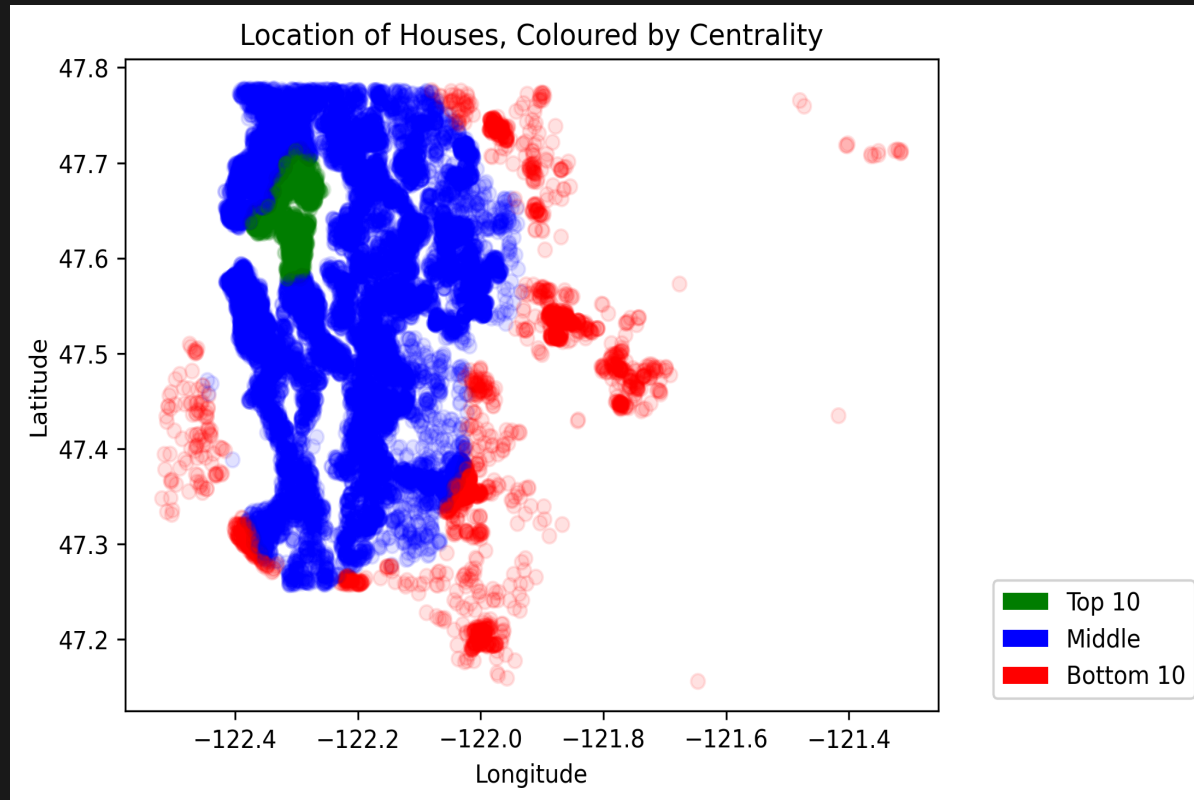
(TOP 10%) CENTRALITY?

A house becomes more central the more neighbours are in a vicinity of fixed size.



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⇒ Sell green and buy red houses.

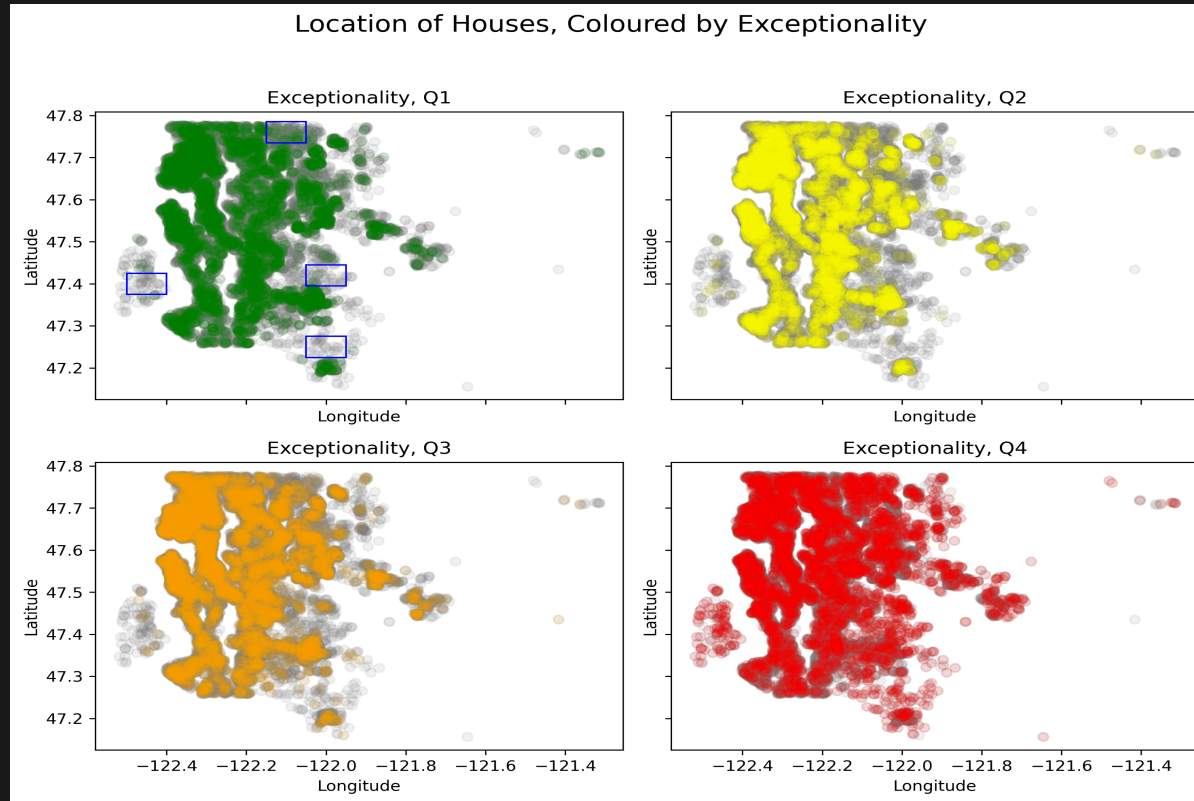
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EXCEPTIONALITY?

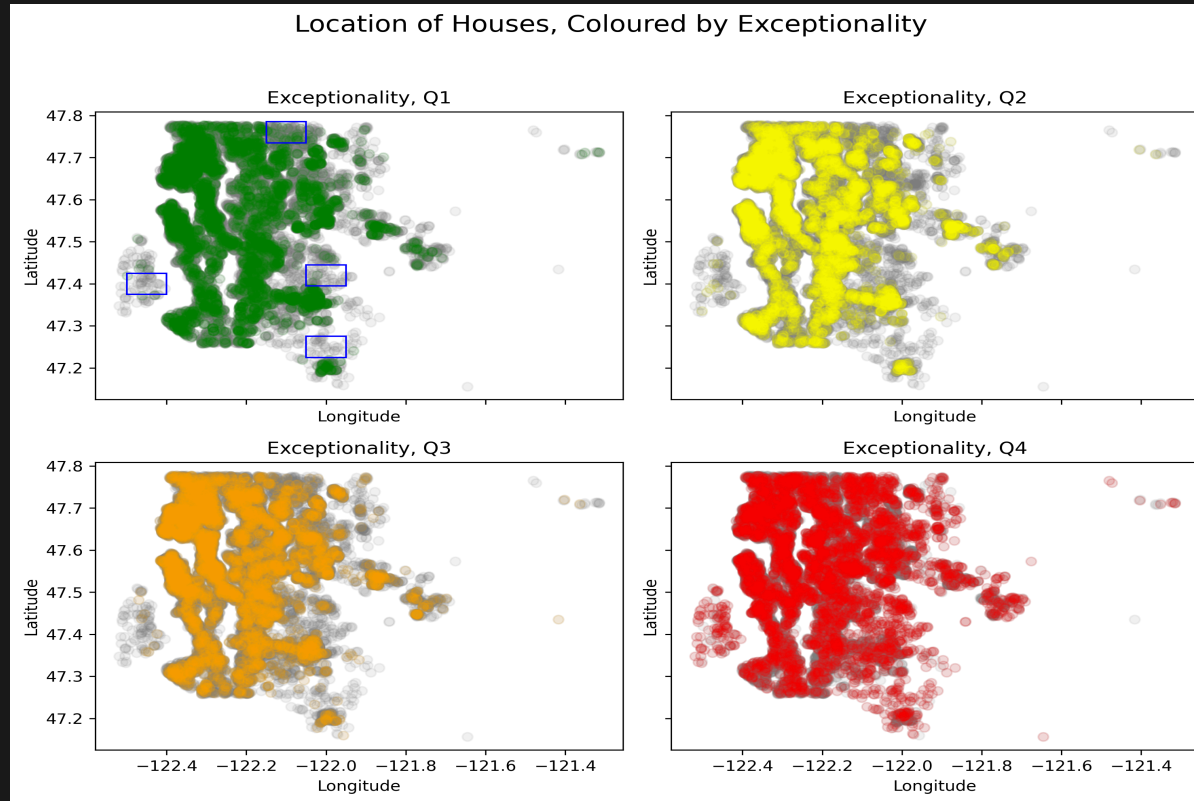
EXCEPTIONALITY?

A house becomes more exceptional the more it differs from its immediate neighbours in terms of living space and lot size.



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⇒ Avoid regions marked in blue.

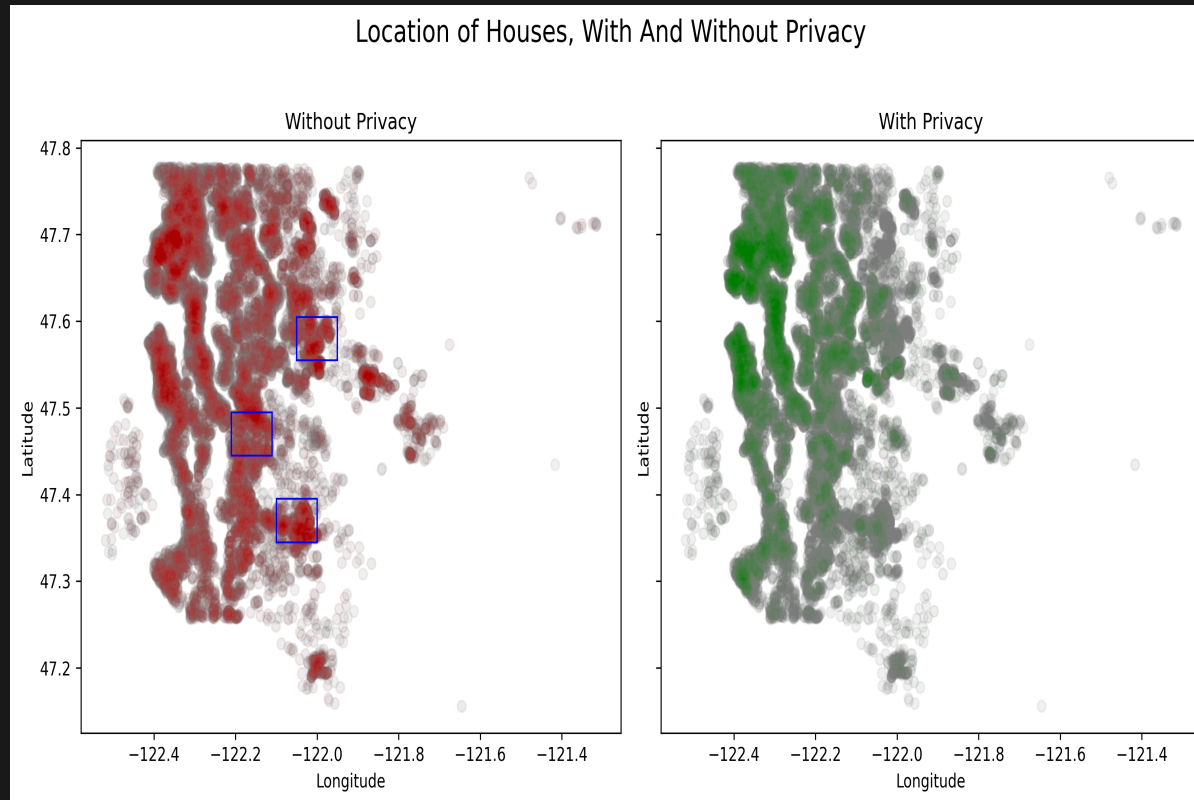
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PRIVACY

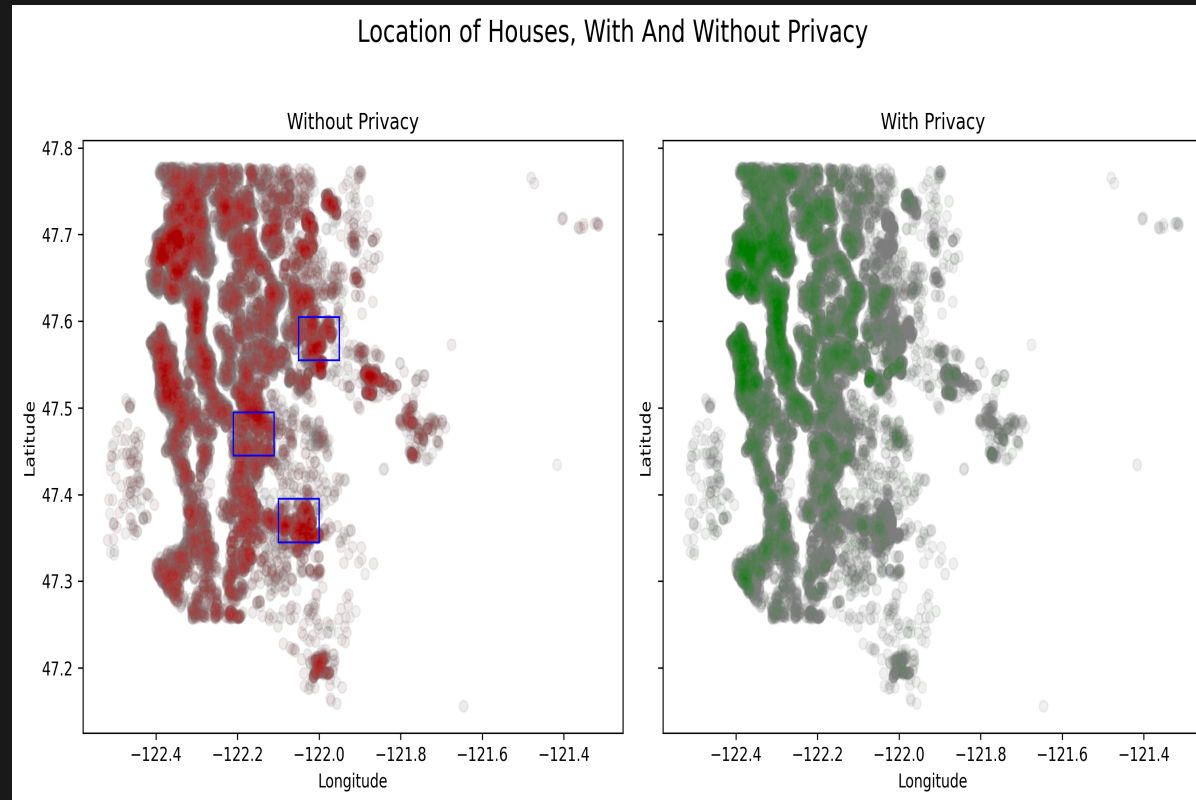
PRIVACY

A house is considered private if and only if it is equipped with a basement.



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THE END.